AMMA: AI-ENABLED MATERNAL MONITORING ASSISTANT

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***Abstract*—Counting kicks by hand is still the standard advice for mothers in the third trimester, and it works reasonably well when the mother can sit quietly and give it her full attention. In practice, of course, attention is exactly the scarce resource: a busy afternoon, a poor night's sleep, or the mother's own movement can all distort the count, and a genuine slowdown in fetal movement may go unnoticed for longer than anyone would like. This paper describes AMMA, a wearable prototype we built over the past year to make that observation less dependent on perception. The belt is centred on an ESP32-S3, which reads an MPU6050 for inertial motion and two piezoelectric discs for abdominal vibration through a 16-bit ADS1115. The same microcontroller drives a small OLED, streams CSV over USB serial, and pushes summary records to a ThingSpeak channel that a companion mobile app renders as charts. Two pieces of logic sit on top of the raw data. A window-based classifier turns one-second IMU windows into posture labels (sitting, standing, walking, lying, moving) shown live on the belt, and a dual-gate rule in the app (piezo spike gate plus maternal-motion gate, with a 1.2 s refractory period) flags candidate movement events whose thresholds can be tuned in a sandbox view. We report here what the bench tests actually showed: which parts of the chain are solid, what the measured signals look like, and where the prototype still falls short. We want to be explicit that AMMA is not yet a validated fetal-kick detector. What it gives us today is a working sensing, display, logging and analysis chain on which kick detection can later be built and clinically evaluated.**

***Index Terms*—wearable sensors, maternal health monitoring, fetal movement, activity recognition, piezoelectric sensor, ESP32-S3, Internet of Things, mobile application**

I. Introduction

Any obstetrician will tell an expectant mother the same thing in the third trimester: pay attention to the baby's movements, and speak up if they slow down. The underlying idea is simple and the supporting evidence goes back decades. Pearson and Weaver's classic Cardiff work formalised it as the "count-to-ten" chart [11], [12], and later studies linked a reduction in perceived movements to stillbirth and other adverse outcomes [13]–[15]. Clinical guidance across the board urges mothers to report reduced movement promptly.

The weakness of the method is not medical but human. It relies entirely on the mother perceiving and counting. A mother who is occupied, anxious, or simply tired can miscount in either direction, and because the counting happens sporadically, a real change in movement pattern may only become apparent after some delay. We started AMMA (AI-Enabled Maternal Monitoring Assistant) with the modest goal of moving part of this burden onto a sensor rather than onto attention.

The concept we settled on has three layers. A soft abdominal belt carries the sensing hardware. Firmware and software then try to tell the mother's own motion apart from events that originate at the abdomen. A companion app finally turns the result into something readable, first for the mother and, if the device ever matures, for her doctor.

What follows is a description of the prototype as it exists today, not as we hope it will eventually be. We focus on what we built and what we measured: the hardware, the firmware, the activity classifier, the cloud and mobile software, and the bench results from both. Where a claim is unproven we say so, because in a maternal-health context an over-claimed result is genuinely worse than a modest one.

The specific contributions of this paper are:

- A multi-sensor wearable platform on a single low-cost microcontroller, combining MPU6050 inertial sensing, two piezoelectric vibration channels through an ADS1115, an OLED display and a wireless data path.
- A hardware-independent, window-based activity classifier running at 100 Hz on the ESP32-S3, with the label displayed live on the belt itself.
- A companion application offering trend charts and a calibration sandbox in which the dual-gate kick-detection rule can be inspected and tuned against streamed data.
- A frank account of the measured piezo and IMU signals, including baseline drift, noise floors and what these imply for any threshold-based detection.

II. Related Work

Fetal movement has been sensed objectively with a surprising variety of hardware, and the literature gave us a useful

map of what had already been tried before we committed to our own design. Zakaria et al. demonstrated that a 3-axis accelerometer alone can recognise categories of fetal activity, which made accelerometry the obvious starting point [1]. Bobrova proposed dedicated signal-processing algorithms for fetal activity in remote monitoring settings [2]. More recent systems have pushed toward complete wearable-plus-cloud pipelines: Jayalath et al. built a sensor-based system covering both fetal and maternal signals [3], and Ansari and Shah described an IoT belt whose stated purpose is tracking kicks in expectant mothers [4]. Xu, Chen and Liu showed that a pair of accelerometers with machine learning can detect fetal movement [5], echoing the earlier wearable textile prototype of Martens et al. [6]. Altini et al. studied the practical trade-off between sensor count and sensor placement for kick detection specifically [6], [16], while Chen et al. explored skin-attachable fibre-optic sensing [7] and Yang and Tavassolian built a simulation system to generate realistic vibrational data for algorithm development [8].

In parallel, and somewhat more mature, is the field of general wearable activity recognition, where inertial features have long been used to classify posture and motion [17]–[20]. That body of work is what justifies our use of the IMU as a maternal-motion context channel.

AMMA's place in this landscape is deliberately conservative. Rather than proposing a novel sensing modality, we combine two established ones (inertial plus piezoelectric) on one inexpensive microcontroller, and we add an on-device activity label so that maternal motion can later be used to gate abdominal events. The piezo channel is the bet; the IMU context is the hedge.

III. System Architecture

Everything in the prototype hangs off a single ESP32-S3 [21]. It does the sensing, drives the display, talks over USB serial and handles WiFi. Fig. 1 shows the assembled board: a perfboard carrying the OLED, the ESP32-S3, the ADS1115 converter and an MPU6050 module, with the two piezoelectric discs mounted on either side.

A. Components

Table I lists the main components and what each one is for.

TABLE I
Hardware Components

Component	Qty	Purpose
ESP32-S3 (N16R8)	1	Controller, USB serial, WiFi/BLE
MPU6050	1–2	Acceleration and rotation sensing
Piezoelectric disc	2	Abdominal vibration (PL, PR)
ADS1115	1	16-bit ADC for piezo signals
SSD1306 OLED (128×64)	1	Local status display

B. Bus Wiring and Addressing

All I^{2}C devices share one bus [22] (SDA on GPIO 8, SCL on GPIO 9), running at 400 kHz; the display is an SSD1306 module [23]. Table II gives the addresses we used. The left

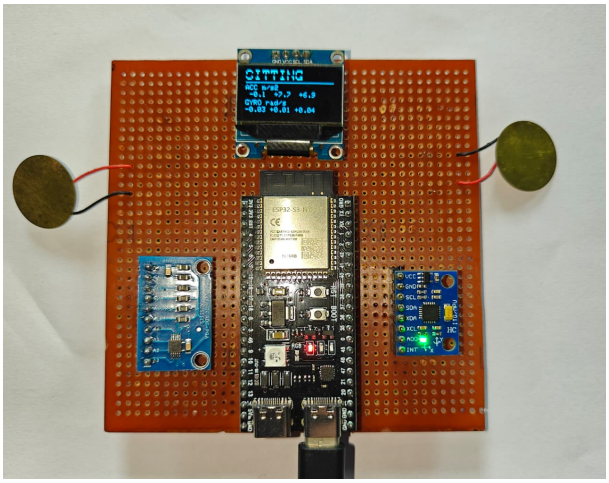


Fig. 1. Assembled AMMA hardware: left and right piezoelectric discs, ADS1115 (left board), ESP32-S3 (centre), OLED display (top) and MPU6050 (right board). The OLED here shows a live "SITTING" classification with accelerometer and gyroscope values.

piezo feeds ADS1115 channel A1 and the right piezo channel A0. The MPU6050's address is set by its AD0 pin, which means a second IMU can join the same bus if we later want one on each side of the abdomen.

TABLE II
I^{2}C Address Map

Device	Address
SSD1306 OLED	0x3C
ADS1115 (PL on A1, PR on A0)	0x48
MPU6050 (primary)	0x68
MPU6050 (secondary, optional)	0x69

C. Wearable Orientation

An accelerometer cannot tell you your posture in absolute terms; it only reports gravity along its own axes. Mount the sensor differently and the same posture gives different numbers, so we fixed a reference mounting before collecting any data: OLED towards the mother's head, left piezo on her left, right piezo on her right, sensor Y pointing towards the head and X towards the left. We record this orientation with every data set because, as Section VI-B shows, forgetting about orientation causes real confusion.

D. Overall Project Flowchart

Fig. 2 lays out the complete flow, from power-up through sensing, classification, display, logging, cloud upload and the mobile application.

IV. Embedded Firmware

A. Acquisition Loop

The firmware samples the IMU on a fixed 10 ms schedule, giving a 100 Hz rate. Early versions reset the timer after each read, and small delays slowly accumulated until the rate drifted; the current code advances the timer by an absolute

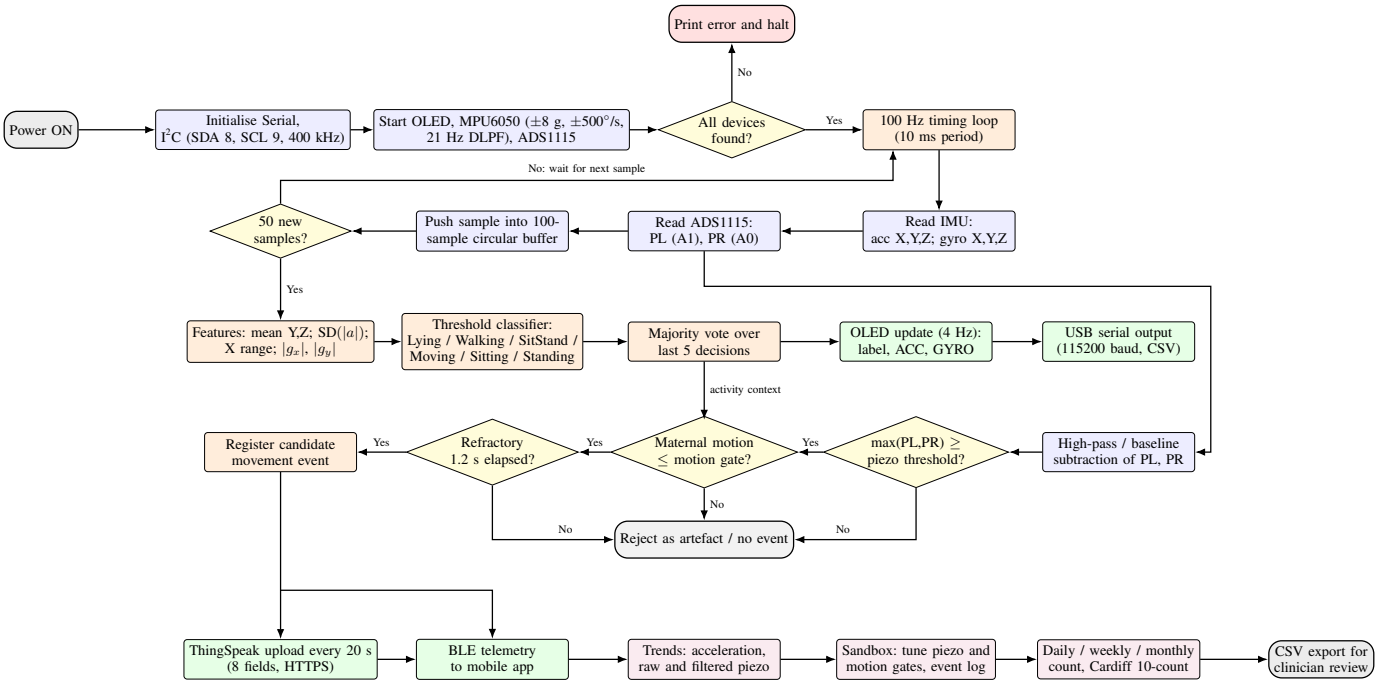


Fig. 2. Complete AMMA project flowchart. Orange blocks run on the ESP32-S3; green blocks are outputs and communication; purple blocks are in the companion application. The piezo gating path (fourth row) is exercised in the application sandbox on streamed data; the activity path (third row) runs on the device.

period instead. Display updates caused a subtler problem. Drawing to the OLED takes about 25 ms over I^{2}C, and since the screen refreshes every 250 ms, the loop used to fall behind and then fire a burst of stale reads to catch up. We now re-align the schedule after each refresh and simply skip the backlog.

The MPU6050 [9] runs with a ± 8 g accelerometer range, ± 500°/s gyroscope range and a 21 Hz digital low-pass filter. The ADS1115 [10] digitises the two piezo channels single-ended, and the raw counts pass through the firmware untouched. That was a deliberate choice: all baseline handling happens explicitly in software where we can see and adjust it, rather than being hidden in the front end.

B. Activity Classifier

The classifier follows the windowed-feature style common in wearable activity recognition [17]–[19], but with one property we found valuable: it has no Arduino dependency at all, so the identical source can be compiled on a laptop and replayed against recorded CSV files. This turned out to matter more than we expected, because it let us debug classification decisions offline against real captures.

A circular buffer holds the latest 100 samples (one second at 100 Hz), and a decision is made every 50 samples. For each window the firmware computes:

- the mean of the Y and Z accelerations, which captures gravity direction;
- the standard deviation of the acceleration magnitude

$$|a| = \sqrt{a_x^2 + a_y^2 + a_z^2}; \tag{1}$$

- the peak-to-peak range of the X acceleration, a proxy for lateral sway;
- the mean absolute gyroscope rates about X and Y.

The decision rules, with the thresholds actually used in the firmware, appear in Table III. Sitting and standing are separated, when the wearer is still, by the pitch angle $\theta = \mathrm{atan2}(\bar{a}_z, \bar{a}_y)$ against a 13° prior, and once a reference posture is known, also by the change in mean a_z across a movement. Because a single noisy window was enough to flicker the label on the OLED, the last five decisions go to a majority vote before anything is displayed.

TABLE III
ACTIVITY DECISION RULES (PER 1 S WINDOW)

Condition	**Label**
mean $a_z > 7.0$ m/s^2	Lying
SD($\lvert a\rvert$) ≥ 0.5 and (X range ≥ 2.3 or $\lvert g_y\rvert > \lvert g_x\rvert$)	Walking
SD($\lvert a\rvert$) ≥ 0.5 otherwise	Sit/Stand transition
$0.12 \leq$ SD($\lvert a\rvert$) < 0.5	Moving
SD($\lvert a\rvert$) < 0.12, pitch $\geq 13°$	Sitting
SD($\lvert a\rvert$) < 0.12, pitch $< 13°$	Standing

A word of caution about these numbers: they come from a small number of recordings made by the project team, under belt tensions and postures we happened to test. They are adequate for the demonstrations in this paper. They have not been validated across different wearers, body types or pregnancy stages, and we would not trust them on a new mounting without recalibration. One label we deliberately dropped during development is “Sleeping”: the classifier can

detect lying still, but lying still is not evidence of sleep, and showing that label felt dishonest.

C. Local Display and Serial Output

The OLED shows the current label in large text, followed by the three accelerations and three angular rates. The idea is that the belt should be legible on its own, without a laptop attached. The same labels go out on the serial port, where a small Python logger on the host writes everything to CSV with an elapsed-time stamp, both piezo channels and the activity label. Every analysis in this paper traces back to those files.

V. Software Platform

A. Cloud Logging

Alongside the fast local loop, the device uploads a summary record to a ThingSpeak channel [24] every 20 seconds. Eight fields carry the raw PL and PR values, the left and right acceleration magnitudes, and the X and Y components of each IMU. The 20-second cadence is a compromise between free-tier rate limits and usefulness: the cloud view is a trend and data-collection record, and nobody should expect to see individual movements in it.

B. Companion Mobile Application

The Flutter application reads the cloud channel and presents five screens: Overview, Trends, Sandbox, Device and About. Overview puts the day's, week's and month's activity next to the Cardiff ten-movement reference. Trends plots the acceleration magnitudes and the piezo signals over a selectable 15, 30 or 60 minute window, either as raw counts or after high-pass filtering (the standard trick for removing the slowly wandering offset typical of piezoelectric elements [25]), with CSV export. The Device screen handles Bluetooth Low Energy discovery and status [26].

C. Dual-Gate Detection Rule and Sandbox

The movement-detection rule implemented in the application requires three conditions to hold together:

$$\text{Event} = \big(\max(P_L, P_R) \geq \theta_p\big) \wedge \big(a_{mat} \leq \theta_m\big) \wedge \big(\Delta t \geq 1.2\,\text{s}\big), \tag{2}$$

where P_L, P_R are the baseline-subtracted piezo values, a_{mat} is the maternal motion measure from the IMUs, θ_p and θ_m are the piezo and motion thresholds, and the last term is a refractory period so that one long roll is not counted five times. A slowly adapting baseline subtracts the large DC offset that belt tension introduces.

The Sandbox exists because we learned the hard way that a single fixed threshold is unreliable; Section VI-E explains why with measured numbers. In the Sandbox a team member can switch between dual-gate, piezo-only and motion-only modes, drag each threshold with a slider (piezo 0–8000 ADC counts, motion 0.05–1.50 m/s^2), and watch the consequences on live data immediately.

VI. Results

A. Hardware Results: Activity Display

The board boots, enumerates every device on the I^{2}C bus, and runs the classifier continuously. Fig. 3 captures the OLED during three conditions we used for acceptance testing. Held still in a seated orientation it reported *SITTING* with $a \approx (-0.1, +7.7, +6.9)$ m/s^2 and near-zero angular rates; carried at normal walking pace it switched to *WALKING*; moved around by hand it showed *MOVING*. Labels, accelerations and rates all remained readable on the 128×64 screen with no overlapping text, which sounds trivial but was not: an earlier layout had the gyro row colliding with the label, and we redrew the screen layout specifically to fix that.

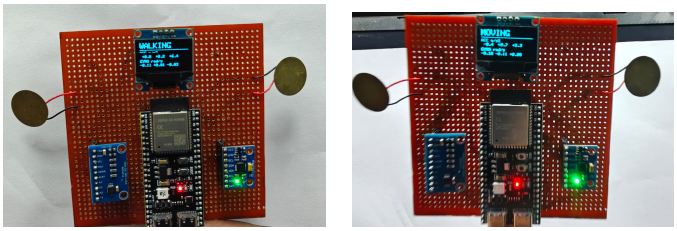


Fig. 3. OLED read-out during walking (left) and general movement (right).

A separate diagnostic mode prints raw sensor values instead of the label. Fig. 4 shows the six IMU channels, the left and right piezo raw counts (PL = 26944, PR = 8176 in this capture) and an "I2C: OK" status line, which confirmed that all four devices coexist on the shared bus without contention.

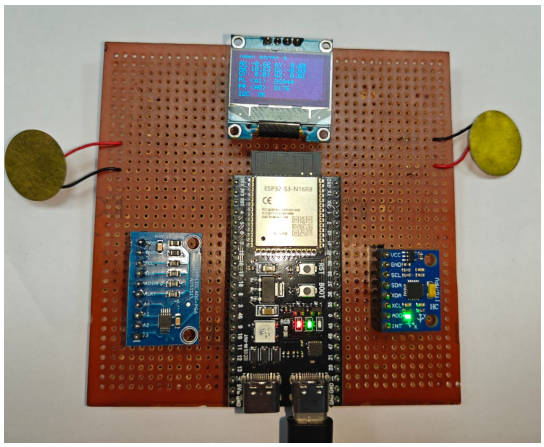


Fig. 4. Diagnostic display showing IMU readings, both piezo channels and bus status.

B. Hardware Results: IMU Behaviour

At rest on a table, the accelerometer reported roughly $(-0.01, -0.03, +10.71)$ m/s^2 and the gyroscope stayed within about 0.03 rad/s of zero, so gravity showed up almost entirely on Z. Worn on the belt in a seated test, the same gravity appeared mainly on negative X at about -9.8 m/s^2. Both readings are correct once the mounting is taken into account, and that is precisely the point: the classifier rules assume

gravity along Y when the wearer is upright and along Z when lying face-up, so any change in belt mounting silently breaks the rules until they are recalibrated.

The two IMUs also disagree with each other slightly. In the app, the left sensor's magnitude trace sits near 10.7 m/s^2 while the right sits near 9.7 m/s^2 (Fig. 5), although both should read about 9.81 m/s^2 at rest. We now apply a per-sensor offset in software before comparing the two, which removes most of the discrepancy.

C. Hardware Results: Piezoelectric Channels

The piezo channels respond clearly to mechanical stimulation, and the numbers are more dramatic than we expected. Tapping the board on a table sent the right channel across a very wide raw-count range, with samples such as $(\mathrm{PL}, \mathrm{PR}) = (592, 23232)$, $(6240, 29392)$ and $(8736, -2576)$. The negative value is worth pausing on: the discs are bipolar devices, and a hard tap can drive the ADC input below its reference. The left channel also moved when we tapped the right side, which initially looked like cross-talk but is really just vibration travelling through the perfboard.

Over longer runs a slower problem appeared: the right channel's resting level drifted upward during one session from about 7500 counts at the start to roughly 29000 at the end. Two practical conclusions follow. First, raw counts can never be compared against a fixed threshold; a baseline has to be tracked and subtracted continuously. Second, and more awkward, piezoelectric elements generate bipolar, high-impedance signals with real voltage spikes [25]. Our current input stage is the discs wired essentially straight into the ADC, which is fine on the bench and unacceptable on a person. Protection, biasing and filtering circuitry are prerequisites for any wearable use.

D. Software Results: Trends and Filtering

Fig. 5 shows the Trends screen. The acceleration chart separates the left and right IMU magnitudes, and the piezo chart toggles between filtered and raw views. In the filtered view (Fig. **??**) the wandering DC level is gone, leaving small transient deflections that peak at about 139 ADC counts in this motionless recording, while the directional acceleration traces stay within a narrow band.

E. Software Results: Sandbox and Threshold Selection

Fig. 6 shows the Sandbox running on the same data. The resting baselines measured about 4041 counts for PL and 5114 for PR, with raw-signal baseline noise reaching roughly 500 counts. After high-pass filtering, the largest transient anywhere in the motionless stream was 138.7 counts. Set against that, the preset "clinical" threshold of 1800 counts is enormous, and the app correctly warns that nothing would be detected unless the slider comes down to about 139 counts or below. For bench work we now use 80–120 counts.

We read this as a genuinely useful negative result rather than a failure. It proves the quiet-state noise floor of the present input stage is low enough to leave working room for

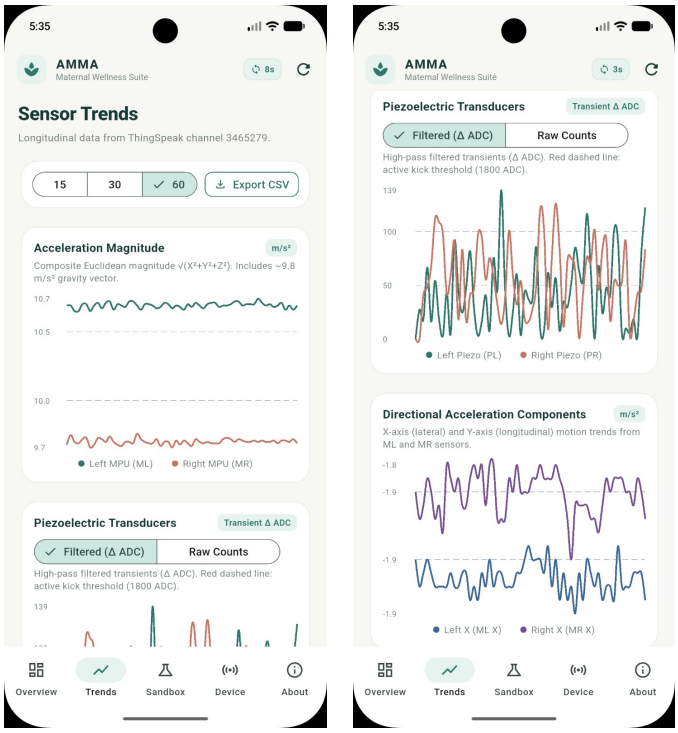


Fig. 5. Application Trends screen: acceleration magnitude for left and right IMUs (left) and filtered piezo transients with directional acceleration (right).

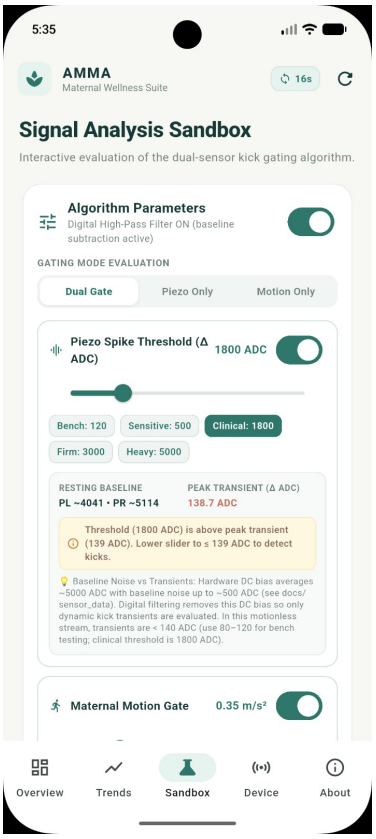


Fig. 6. Signal Analysis Sandbox with gating-mode selector, piezo threshold presets (120 to 5000 ADC counts), measured baseline and peak transient, and the maternal motion gate.

a threshold, and it proves just as clearly that no motionless recording can tell us where a real-kick threshold should sit. That number can only come from labelled recordings of actual movements, which we have not yet made.

F. Summary of Verified Behaviour

Table IV collects what is demonstrated and what is not.

TABLE IV
SUMMARY OF PROTOTYPE STATUS

Capability	Status
Shared I^2C bus with OLED, IMU, ADC	Working; "I2C: OK" confirmed
100 Hz IMU sampling	Working
OLED activity display	Working for sitting, walking, moving
Piezo response to tapping	Observed on both channels
Cloud upload and charts	Working (trend view, 20 s interval)
Mobile trends and sandbox	Working on recorded/streamed data
Activity thresholds	Preliminary; need recalibration per mounting
Serial-to-CSV logger	Port detected; full five-minute capture to be re-verified
Kick detection accuracy	Not yet evaluated
Clinical validation	Not done

VII. Discussion and Limitations

If this project taught us one thing, it is that reading the sensors was never the hard part. Deciding what the readings mean is. Walking, breathing, coughing, clothing friction and posture changes all generate vibrations that, at the raw signal level, look uncomfortably like a movement under the skin. That observation shaped the whole design: it is why a separate maternal-motion channel exists, why a refractory period exists, and why we treat the on-device activity label as context for gating rather than as a product feature in itself.

Several limitations deserve plain statement. The activity thresholds rest on a handful of recordings from a few people. The piezo front end is a bench arrangement with visible baseline drift. The IMUs need individual calibration. Most importantly, no recording has ever been labelled against an independent reference for fetal movement, so the sensitivity and false-alarm rate of the kick rule are simply unknown. And the device is an engineering prototype: it must never be used to judge fetal wellbeing or as a substitute for normal medical assessment.

VIII. Conclusion and Future Work

AMMA, as presented here, is a wearable maternal-monitoring prototype that fuses inertial and piezoelectric sensing on an ESP32-S3, shows a live activity label on an OLED, logs over serial and cloud, and ships a mobile application with trend charts and a calibration sandbox for a dual-gate movement rule. The bench results support a straightforward conclusion: the sensing, display and logging chain is dependable, the piezo channels respond measurably to stimulation, and the limits on threshold selection and calibration are now well understood rather than guessed.

The path forward is equally clear. We need a protected, biased analogue front end for the piezo discs; labelled data sets covering each posture and deliberate abdominal stimulation across several volunteers; a recalibrated classifier for the final belt orientation; the dual-gate rule ported onto the ESP32-S3; and lightweight machine-learning models trained and tested with subject-independent evaluation. Clinical assessment only becomes meaningful after those steps, and only with ethical approval and clinical partners in place.

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